

home network from a DHS device to a DHR device. The digital home media format model defines a set of required media formats and a set of optional media formats for each of the three classes of media: imaging, audio, and video. The network interoperability model for media formats is as follows:

- m All DHS devices and DHR devices must support all formats as required for any of the media classes they support. All DHS devices and DHR devices may support any additional formats designated as optional for any of the media classes they support.
- m Any DHR device must be able to receive content from any DHS device. A DHS device may stream content in its native format if the receiving DHR device supports such native format. If the DHR device does not support the content's native format, the DHS device must transcode the native format to one of the applicable required formats, or to a format understood by the rendering device. Interoperability for audio devices in the digital home is ensured through the requirement to support the LPCM audio format. LPCM is an important, uncompressed audio format widely used today for the interchange of single and multi-channel premium-quality audio streams between digitally interconnected devices. Direct transmission of LPCM between devices is the means for ensuring the highest fidelity of premium audio content. The technical requirements to create an LPCM stream from any source content and to transmit that stream are well within the capabilities of present day and future devices and digital interconnects. As a required format in the digital home, LPCM ensures the broadest range of interoperability with the lowest possible cost and complexity.

LPCM represents a reasonable technical choice for a required audio format, particularly in wired environments. Wireless networking is rapidly growing in importance for home networking and is expected to become an important means of distributing media in the home. For a wireless device, or a resource-constrained device such as a portable player with limited memory and power supply, compressed audio formats in the optional format set provide more efficient usage of network bandwidth, battery power, and storage. Vendors whose products fall into these classes should seriously consider supporting one or more of the optional compressed audio for-